

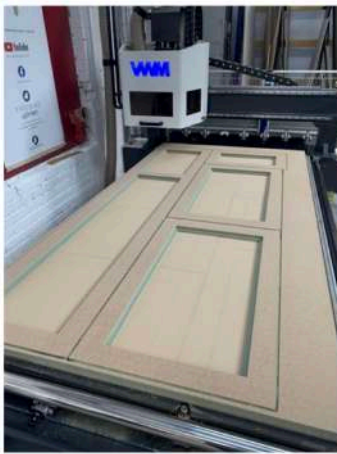
Trade Shaker Door Interface Guidelines

*Prepared by Alastair Johnson for Superdesign.tech, 8th April 2024
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teams.*



FREEBIRD TRADE SHAKER DOORS

Flat, square, stable MDF doors for fitted furniture makers.



Scope

The purpose of this document is to clarify current requirements for a web and mobile-based interface to allow our customers to specify and order shaker style doors. The interface should be mobile-friendly, visually consistent between mobile and desktop. The interface should work for individual door specification and also for 'catalogue mode' repeat quantity orders.

This document should be read now by the whole development team, then used as an ongoing reference. It includes all current required specifications, plus future desirable features which should be taken into consideration when making decisions on the user interface.

I am including suggestions on how the user interface might look, but I fully expect feedback and changes to these suggestions.

Initial Launch

The deadline for making the system live is 18th May when I will be publicising it at 2 shows in the Birmingham National Exhibition Centre:
<https://www.makerscentral.co.uk/>
<https://www.materialsandfinishesshow.com/>

Target Customer

At this stage, the target customer is:

Small to medium bespoke fitted furniture makers in the UK.

Target Usage

Cabinet-style doors to be applied to made-to-measure cabinetry, where frontages are primarily in a shaker style and will ultimately be painted and will be mounted on concealed sprung hinges (ie Blum): Wardrobes, kitchens, alcove cabinets, large display cabinets, understairs storage: especially including angled work and unique door designs such as multi-panelled doors and varying panel styles including reeded panels.

These doors will most often be supplied as raw MDF for the customer to finish themselves.

Brief Competitor Analysis and Our USPs

Here are some examples of competitors in this product space targeting the same type off trade customer.

Large specialised setups:

Please review these links for inspiration on ways of presenting (and improving!) the ordering interface:

<https://cworkshop.co.uk/products/made-to-measure-shaker-doors/>
<https://www.thedrawerboxcompany.co.uk/product/shaker-door/>
<https://www.proboxdrawers.co.uk/category/shaker-doors/>
<https://www.proboxdrawers.co.uk/product/shaker-kitchen-cabinet-doors/>
<https://ktechcnc.co.uk/>
<https://thecabinetshop.co.uk/products/bspoke-doors>
<https://bespokedoorstudio.co.uk/>
<https://doors2size.co.uk/>

Small CNC workshops filling in between their own bespoke work:

Another type of competitor are other bespoke furniture makers with their own CNC machines who are filling in between their own jobs by providing doors to local makers. Some of these are potential partners for us.

<https://creativecabinetry.co.uk/services/>
<https://www.wardrobe-doors.com/collections/shaker>

In general the current level of online customisation and visual feedback is very limited from our competitors

Future Potential Customers and Uses

- People seeking 'Ikea hack' components (catalogue of special doors made to suit known cabinets like Ikea Pax - especially high-end styles like fretwork).
- Bigger fitted furniture making companies looking to outsource to increase their productivity
- Kitchen Distributors (Doors fully finished)
- DIYers building their own custom cabinets, or even for wall panelling.

Future Competitors

When our door builder starts to target DIYers / Homeowners too, we will be competing with suppliers like these:

- <https://shaker2go.co.uk/>
- <https://fittedwardrobedoors.co.uk/>
- <https://justwardrobedoors.co.uk/>
- <https://bespokedoorstudio.co.uk/product-category/ikea-doors/>

Features and Benefits

The distinctive features and why they benefit our target customer:

Feature	Benefit
Outsourced	Although our target customer is capable of manufacturing doors themselves, outsourcing frees up space and time in their often cramped workshop, allowing them to focus on other parts of the project, and to increase their productivity while keeping overheads low.
Reliably Square and Precisely Sized	Gets rid of common pain-points experienced with self-made 5-piece doors, enabling a higher quality final fit and finish, with precise door gaps, minimising re-work time and increasing end-user satisfaction and business reputation.
Supplied as Raw MDF Components	By focusing only on what the CNC does best, we minimise their costs and allow them to control how they finish the parts, making the final doors their own.
Solid MDF Frame Construction (ie panel openings are created in a subtractive way by through-cutting a door-sized board, instead of jointing rails and stiles together.	Smooth, no joints, enables a higher quality final finish, and no cracks opening up, and stronger than any possible jointed option. Also creates design freedom - with angles and multiple panels more easily achieved.
Separate Panels ('2 piece construction')	Whereas some MDF door manufactures use '1-piece' construction which requires sanding of a recessed panel, we drop in a thinner board, ready for finishing. This also creates a more stable door construction (less prone to warping). Crucially also, our construction method allows the insertion of different panel styles of varying thicknesses (reeded; fretwork; fabrics; melamine effects), to be accommodated in a standard rebate style within the same door construction. This gives our customers design freedom and the ability to upsell new styles to their own customers.
Online Visual Door Builder and Estimator	Our customers often put together quotes and designs outside of normal working hours, when we are not contactable. The planned ordering system allows them to control their costs and give their customers accurate quotes, without waiting on our prices, or working out how long it might take them to make an unusual door themselves. This gives design freedom and wins back their admin time, plus giving confidence that we will supply exactly what they need without error, and keeping their costs low because our own admin time is minimised



Our customer Alex Burgess

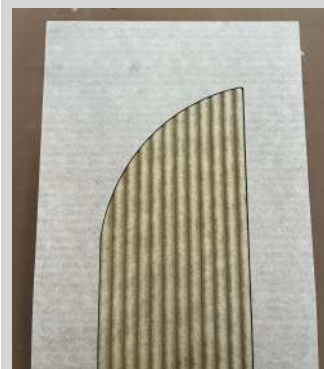


Mark Upperdine has a large workshop and spray area but relies on us for door manufacture to spend more time on site and reduce his customers' waiting times.

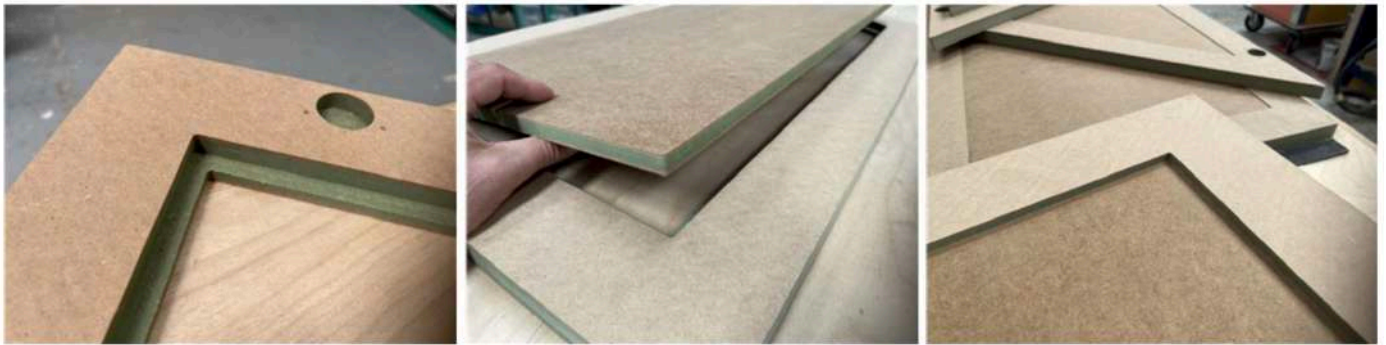
Future Benefits

As the customer base is widened out in future, we will add the option for fully finished doors which will be a benefit for direct-to-consumer sales.

The solid frame construction method means we could easily manufacture arches or other special shapes in future too, without changing the basic manufacturing process.



Specifications



Refer to this section for how the doors need to be set up.

Primary Material

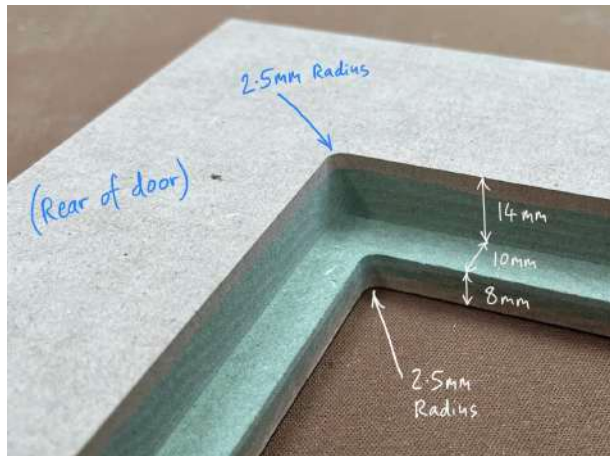
Finsa Fibrapan Hidrofugo (High Grade Moisture Resistant MDF).

Door Thickness Options

22mm is the standard shaker door thickness; this can be used with reeded panels and fretwork options. Can also be selected for slab doors, cover panels etc .

18mm currently this will only be offered for slab doors, plinths and cover panels that may also be ordered through the door builder system.

Standard Door Rebate



The 22mm door material is through-cut and rebated from the back of the door, using a 12mm diameter cutter, finishing off with a 5mm diameter cutter, so the internal corners have a 2.5mm radius. The rebate is always 10mm wide, and 14mm deep, leaving 8mm at the front.

Standard Panel Options

Standard panel: 12mm thick Fibrapan Hidrofugo (for 22mm thick doors). These creates a robust-feeling door, where the panel is still expressed at the rear of the door with a 2mm inset that allows for a small arris rounder of the surrounding opening, but by hiding most of the rebate, it reduces finishing work.

We may add an 18mm shaker option in future, with different standard rebate and panel thickness, but we would advise customers to choose 22mm for tall doors.

25mm may also be offered in future, where a more robust door is required, or when a deeper front recess is desired in front of reeded panels or 18mm melamine panels.

We may in future offer a smaller diameter cutter for internal corners.

If we offer 18mm and 25mm thick doors in future, the rebate depths may be different.

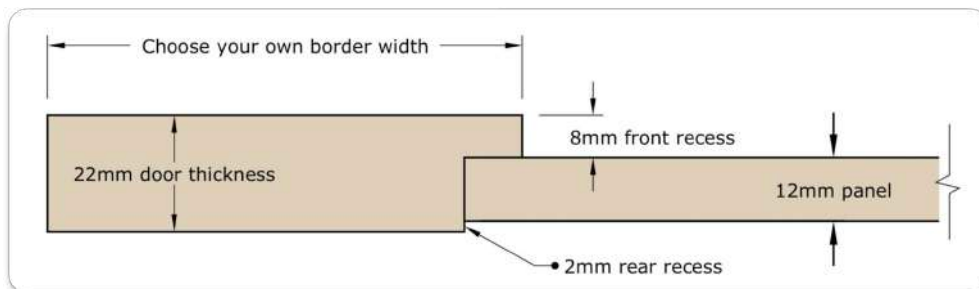
We may in future give the option of choosing 6mm or 9mm panels. Some customers prefer thinner panels with a deeper rear recess on the doors. And it will be necessary to use thinner panels if we offer 18mm shaker doors .

25mm doors may also be offered in future, where a more robust door is required, or when a deeper front recess is desired in front of reeded panels or 18mm melamine panels.

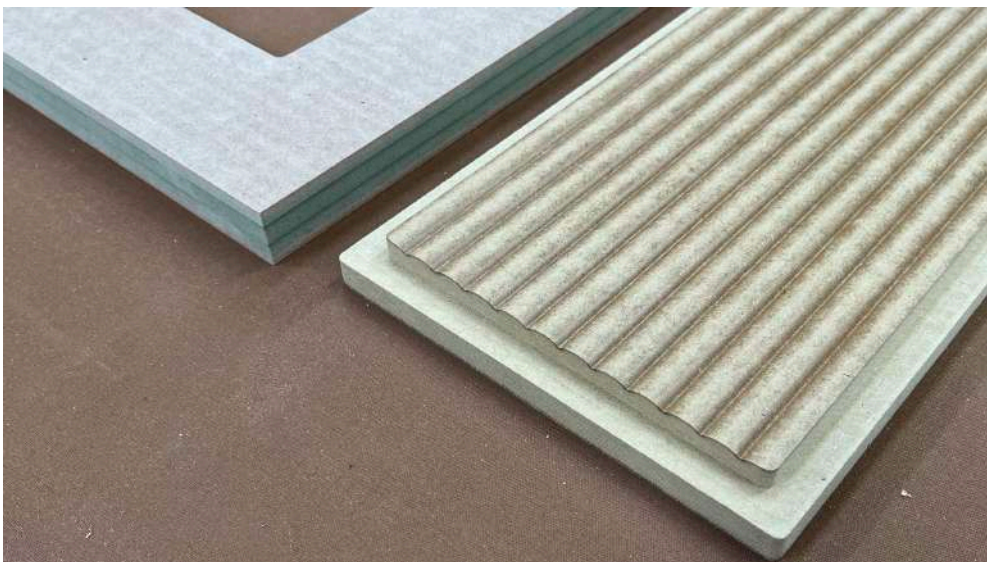
The image below shows the back of a standard door with a 12mm panel, prior to any finishing work:



The diagram below shows the standard door in section

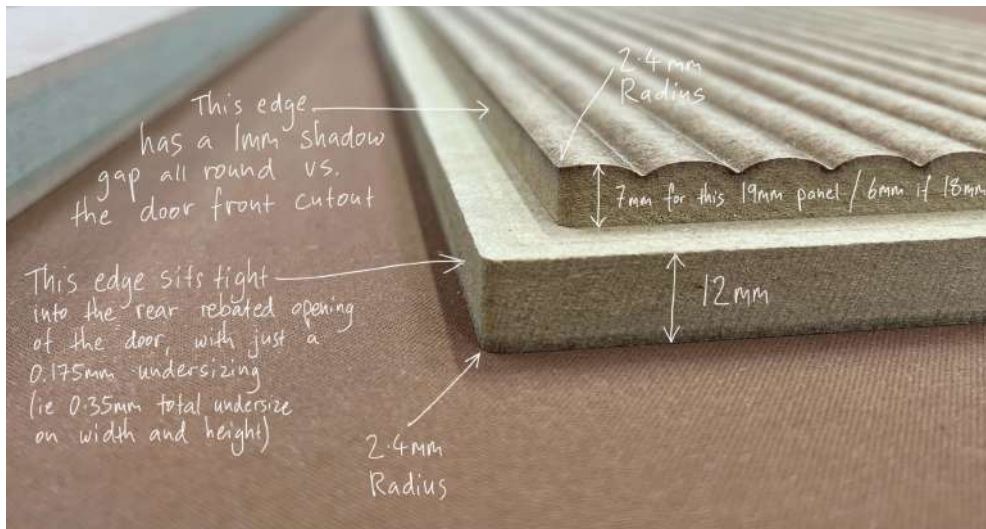


Reeded option: 19mm thick Finsa 'Tex Flute' (for 22mm thick doors only). This will be machined face up on the CNC, to create a mating rebate with the door frame, as described in the images below:



The system should also allow for glass door panels in future, but probably it will be up to the customer to supply their own glass. Probably we will supply a rear retaining piece in thin MDF. Aiming to keep the standard door rebate the same. TBC





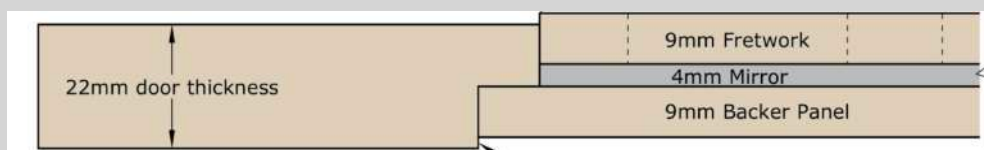
18mm Melamine Option: This will typically be used for a fabric-effect board such as Finsa's 'Canvas Greige' - see below where this has been paired with a reeded panel, shown before and after spraying:



The machining settings for the 18 melamine panel should provisionally be the same as the 19mm reeded panel, as sketched on the photo above, however it is very important that we have the option to adjust tolerances in the back end of the system, such as the undersizing of panels to ensure best fit, as we may still need to do further refinement on this for different materials and uses. The same is true for radiuses.

High Priority Phase 2 Feature:

We will definitely want to offer fretwork door panels, which can be introduced progressively as each pattern variation is programmed and tested parametrically. This will become one of the biggest selling points of the system.

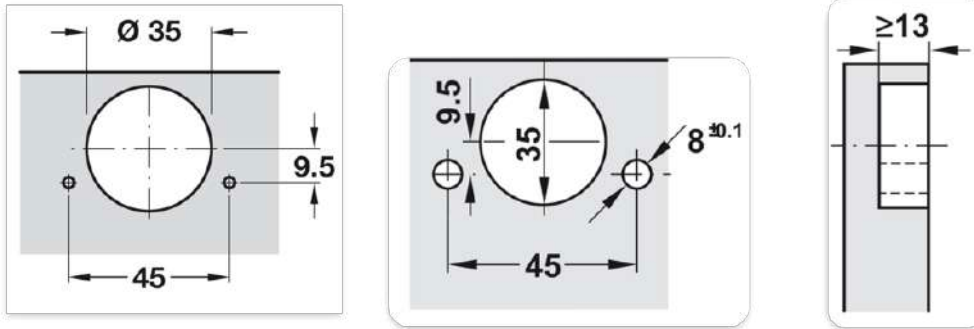


Hinge Drilling

Cup hole: 35mm $\varnothing$, 13mm deep, 5mm gap to door edge (ie 22.5mm to hole centre), to be drilled with a 35mm drill, not pocketed

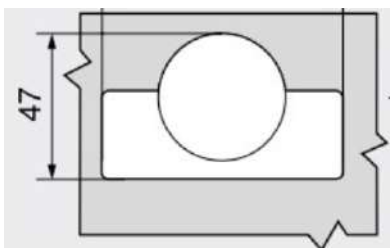
Standard option: screw point holes to be drilled 0.5mm deep using a 4mm $\varnothing$ v-tipped drill.

Alternative option: 8mm $\varnothing$ holes 13mm deep for blum 'inserta' style hinge fittings.



Minimums

Borders with hinges: When hinges are added, the minimum border width is 65mm (measured at front face of door). This is to ensure there is space for the maximum hinge boss fixing plate as shown below for a Blum 170° hinge. The calculation is based on the hinge and fixing plate dimension of 47mm, plus 5mm gap off door edge, plus 3mm clearance, plus 10mm wide rebate = 65mm. *(Note that for most types of hinge the boss dimension is actually smaller than this but we will stick with 65mm as the all purpose standard minimum for any hinge side, for strength purposes also.)*

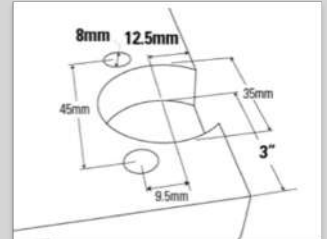


Other Borders: Where there are no hinges on an edge, the system should allow rails or stiles to be thinner, down to a minimum of 35mm.

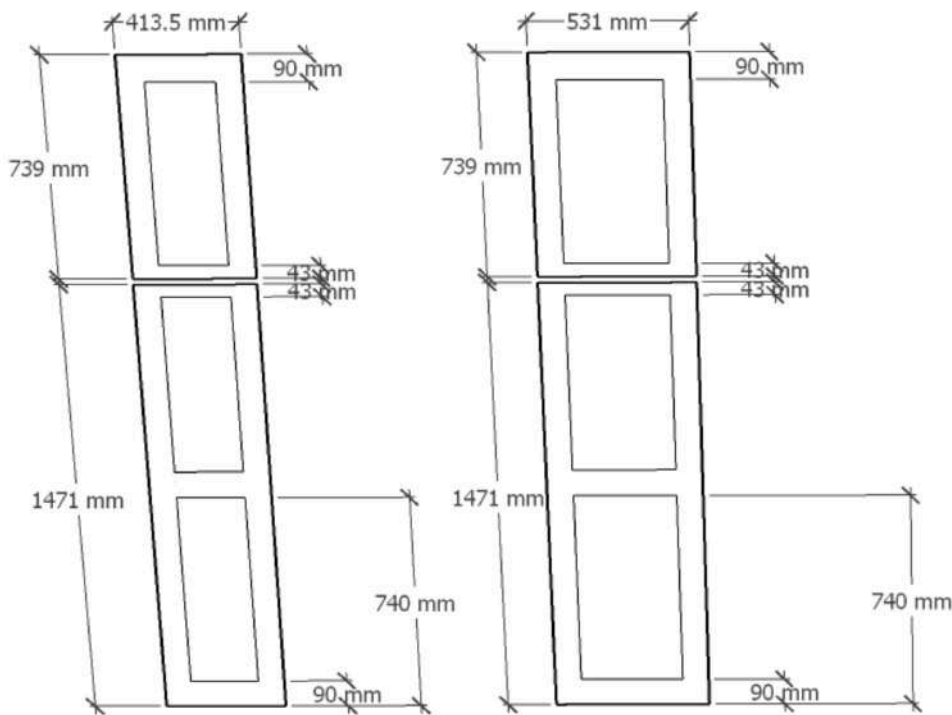
Here is an example use-case:

Our trade customer Dan Fyfe recently required tall doors with 3 panels and 90mm borders, but in one location he wanted to split the upper panels into separate top doors, while keeping the rails in line with adjacent full doors. So, accounting for 4mm door gaps, he specified those rails at 43mm. This is a common requirement for our target customer, so the system must allow for it.

We may add special hinges variations in future: 26mm small hinge hole / 40mm heavy duty hinge hole (eg Sugatsune J95 hinge) / 35mm overlapping edge hole (Eg for Sugatsune ADH-P flap hinges) but this is not a high priority.



In future we would like to sell the hinges and mounting plates themselves via the same ordering interface.



Finishing Options

1: Raw Unassembled: Parts will be supplied straight off the CNC machine, with panels placed loose into their rebates for transport, but it will be up to the customer to do any finishing work on inner edges; glue in the panels; and to do any outer edge finishing work, and caulking of panel gaps, before painting. Customers have the option of manually squaring off the front internal corners if they prefer that look.

2: Assembled and Prepped: We will do the arris rounding, edge sanding, gluing in of panels, and caulking of gaps. We will offer this in one standard way only - with 2mm arris roundovers all round, and leaving the 2.5mm inner radiuses as they are, then edges sanded to 240 grit. We do not offer edgebanded edges.

In future the system will include fully finished doors. This will be when it becomes more aimed at end users. We are avoiding offering this until all other issues are ironed out, as the risk of damage in transport or while installing can cause us costly liability issues.



3: Primed: For now we may offer this but we will price it highly as a disincentive, and we will emphasise that we cannot guarantee adhesion of

the customer's choice of paint to our primer (we will use Sayerlack AU474 waterbased primer)

Pricing

All of the following is provisional and may be subject to review, so ideally the pricing formulas should be accessible for us to change in the back end of the system

Here is the formula currently used for pricing, excluding VAT

Fixed fee per door: £3

PLUS

Total door area in sq m x £75

(This covers the 22mm and 12mm panels. For angled doors, the lost corner is ignored in the calculation - it is still priced as if rectangular)

PLUS

£25 fixed fee if the door is angled in any way

PLUS

£5 per added mid rail

PLUS

£1.50 per hinge hole

Slab options - 18mm and 22mm

Currently these are priced the same as above, but using £45 instead of £75 per square metre. **This is provisional and may be reviewed**

Panel Upgrades

Formulas above to remain the same,

PLUS

£10 fixed fee per special panel

PLUS

£60 per square metre of *panel material* for reeded panels

OR £40 per square metre of *panel material* for melamine panels

This is provisional and may be reviewed

Delivery

Fixed fee for organising delivery: £20

PLUS

£0.45 per mile (calculated as the round trip distance)

PLUS

£13 per hour inclusive of offloading

CNC Machining

Refer to this section for how the doors will be machined by an 8-tool changer 8'x4' flatbed CNC machine.



Standard Tooling:

Our standard tooling is as follows, in the order we have them loaded in our tool changer

T1 (Tool position 1): 4mmø V-point drill

For marking screw points for hinges, 0.5mm deep

T2: 5mmø brad point drill

- not used for shaker doors -

T3: 5mmø straight cutter

For final perimeter and rebate cuts to leave 2.5mm radius. Also for pocketing out the 8mmø inserts fixing holes

T4: 8mmø compression cutter

For general profile cuts

T5: V-cutter

- not currently used for shaker doors. But for fretwork panels this tool position may be used, with [this tool](#), for adding a slight rounded chamfer to edges to aid the finishing process. -

T6: 12mmø straight cutter

For rebating, and inner opening cutting leaving and onion skin.

T7: 15mmø brad point drill

- not used for shaker doors -

T8: 35mmø flat bottomed drill

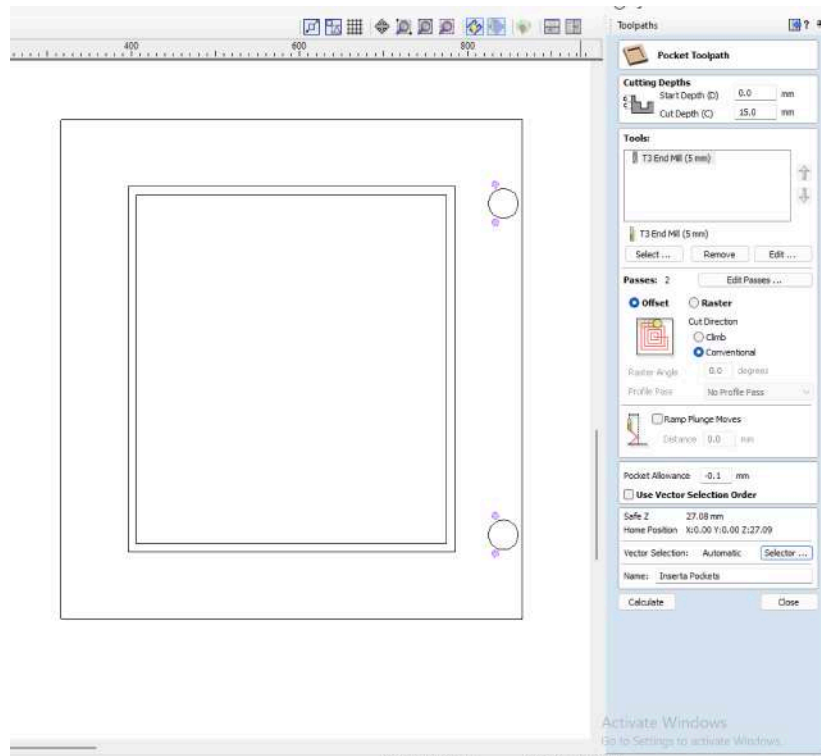
For hinge cut hole cutting

We may in future offer a smaller diameter cutter for internal corners.

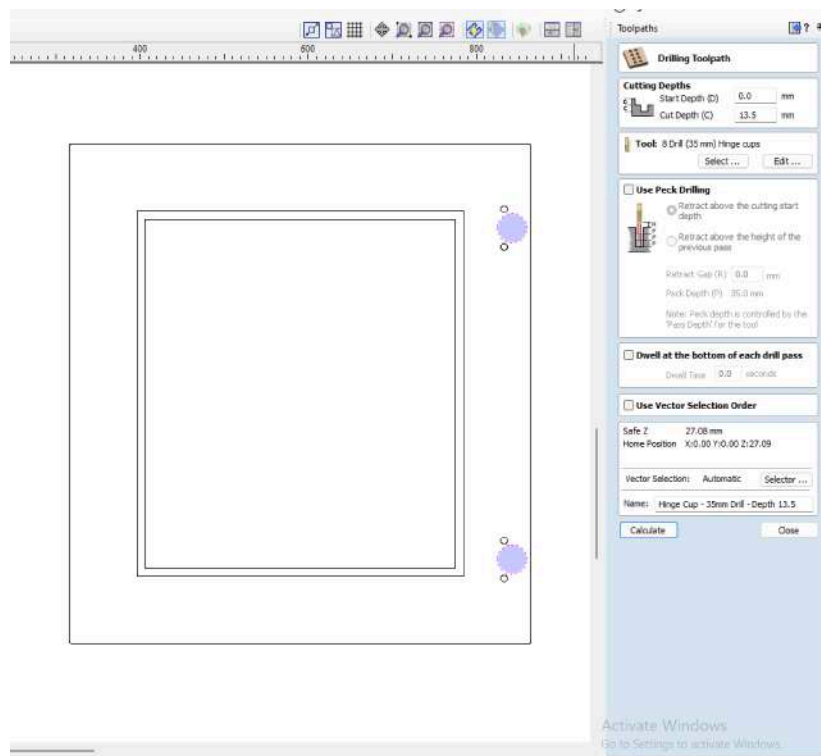
Toolpaths and Order of Cuts

Screenshots from Vectric, showing the order of operations, designed to minimise movement of parts. 22mm doors.

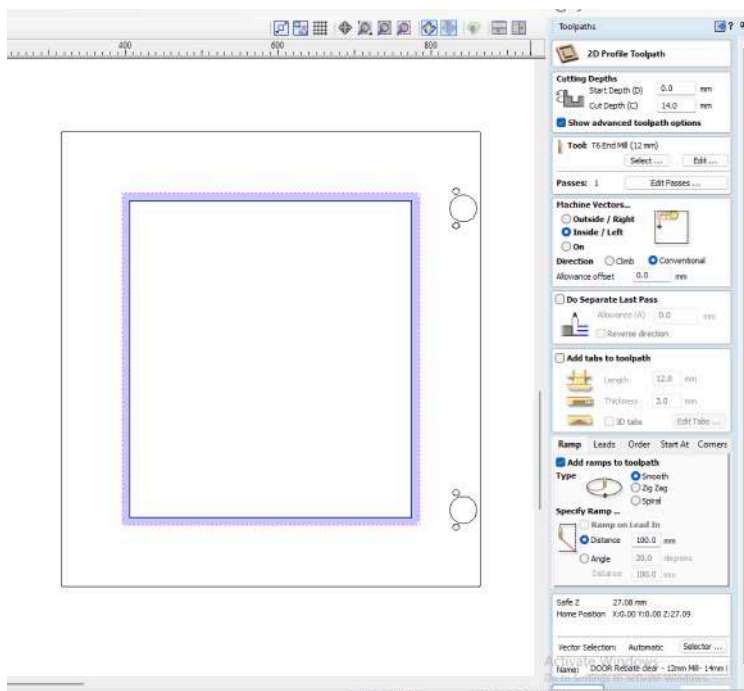
1: Hinge fixing holes, either screw points or inserta recesses as shown:



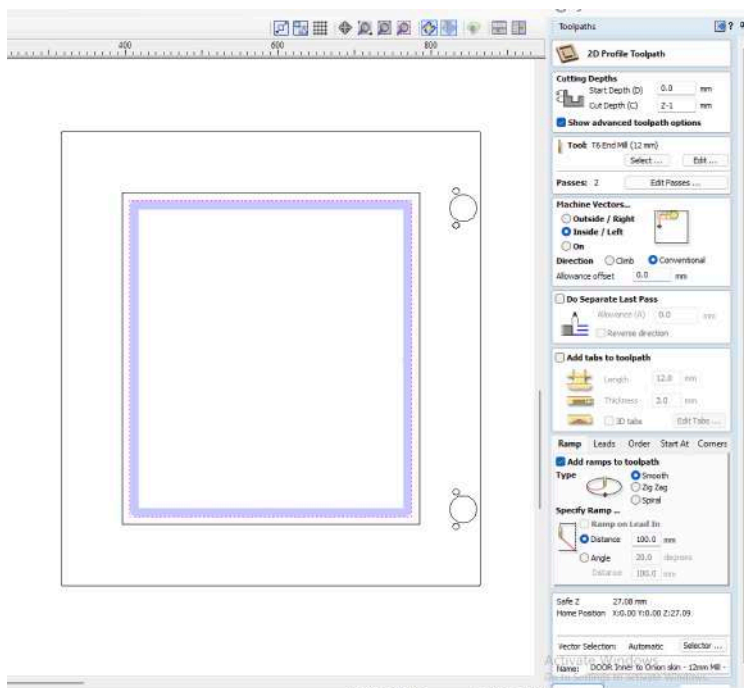
2: Hinge cup holes:



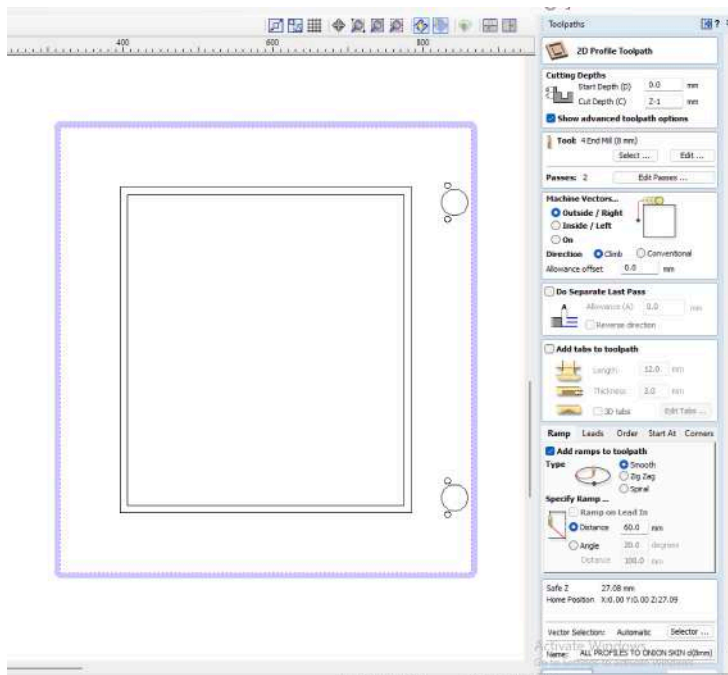
3: Inner Rebate (12mmø cutter)



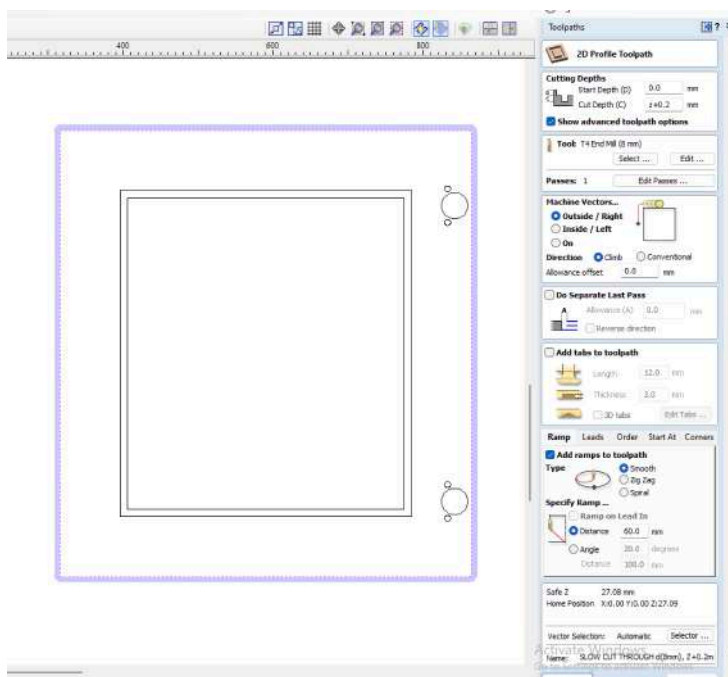
4: Inner perimeter cut (12mmø cutter) leaving a 1mm onion skin



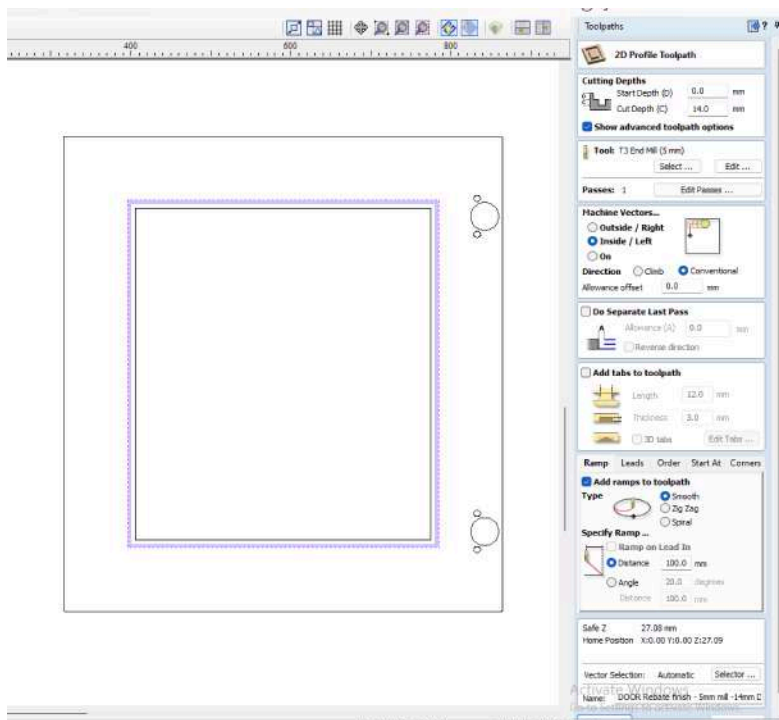
5: Perimeter cut (8mmØ compression cutter leaving a 1mm onion skin).



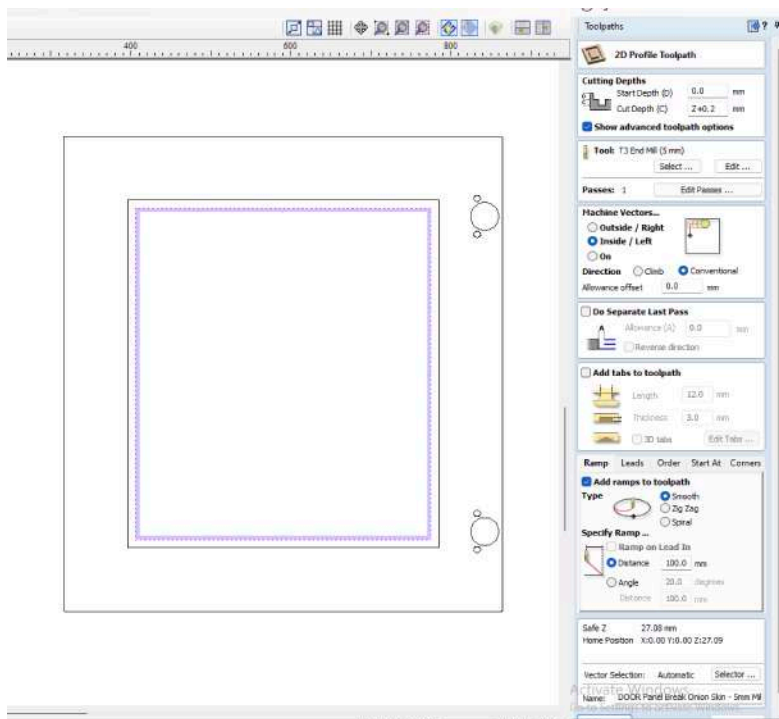
6: Perimeter through-cut (8mmØ compression cutter)



7: Rebate finishing cut (5mmø straight cutter - to reduce corner radius)



8: Final inner through-cut breaking onion skin (5mmø cutter - to reduce corner radius)

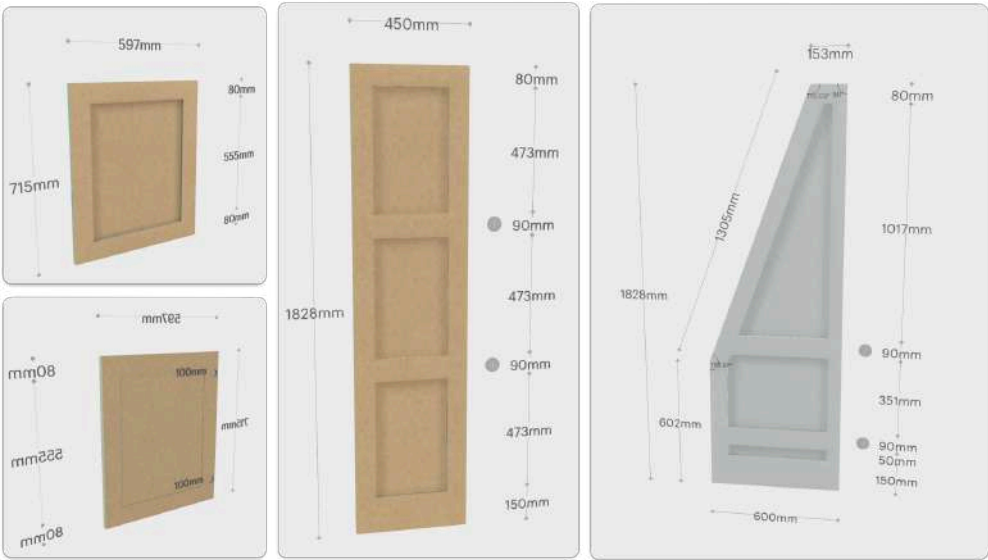


Then 12mm panel: simple one pass perimeter through-cut with 8mm compression cutter.

User Interface Suggestions

Please design the interface with mobile in mind. There are 2 main factors to consider

1: Designing an Individual Door: This interface should centre around a visual representation of the door, so all settings that have been chosen are reflected in a 3D image of the door to avoid error. The current 3D door display is good as per these examples - all key dimensions are clearly represented.



2: Compiling an order of multiple doors: let's call this 'catalogue mode'. This must be a separate interface that allows users to see all key dimensions and specs in a simple text readout. It should also be easy to click back to the 3D editor for any of the doors in the list, for more detailed editing, however it must also be possible to directly edit these text fields without having to jump back to the 3D door editor for each door.

	Style	Finish	Description	Price £	Qty	Subtotal
	 Shaker Door 22mm / 12mm Panel	Raw - Unassembled	715 x 577mm Border: Equal - 80mm Mid Rails: None Hinge Drilling: Left - T100 B100	£38.00	2	£76.00
	 Shaker Door 22mm / 6mm Panel	Assembled and Prepped	1828 x 430mm Border: Custom - L80 T80 R80 B150 Mid Rails: 1: 623(90) 2: 1186(90) Hinge Drilling: Right - T100 T643 B100 B643	£110.57	4	£442.28
	 Slab Door 22mm	Primed	1987 x 719mm ANGLED HL-1708 TWL-0 HR-1987 TWR-480 Hinge Drilling: NONE	£179.83	1	£179.83
	 Shaker Door 22mm / 12mm Panel	Finish Painted F&B 'Lamp Room Grey no. 88'	1828 x 600mm ANGLED HL-602 TWL-0 HR-1828 TWR-153 Border: Custom - L80 A80 T80 R60 B150 Mid Rails: 1: 200(90) 2: 641(90) Hinge Drilling: None	£220.50	1	£220.50
						Total Goods: £918.61
						Delivery Cost: £0.00
						VAT: £183.72
						Total Owed: £1,202.33

For example imagine that a customer has set up their preferred standard wardrobe door style: the top rail and stiles are all 90mm, the bottom one is 120mm, and there is a mid rail 150mm tall which should always be 800mm from the bottom of the door. Hinges have been set to be 100mm from top and bottom of the door, with 2 additional hinges evenly spaced between.

With these settings established the user will typically want to perform the following actions quickly within catalogue mode:

- **Duplicate a door** (perhaps either right click and select 'duplicate', or perhaps there is a round '+' button or similar next to each line item to make this easier.
- **Switch the hinged side** - it should be easy to swap from left to right hinge side drilling, while retaining the hinge spacing and other attributes of the door. This will be a very common operation for example for a run of wardrobe doors like this:



- **Change height and width**
- **Change level of finish** - with this and other changes they may want to just quickly see the impact on the total cost of the order to help them decide what to choose, and there is no need for them to do this via the 3D interface
- **Change quantity** of an identical line item, for multiple doors the same.

Sometimes they may want to duplicate a line item to give them a head start on editing, and then they will want to go back into the 3D editor to make more detailed changes. For example for the design pictured at right, they may have set up all the tall doors, then use a duplicate to go in and remove the mid rail and reduce the height and hinge quantity to create then short central doors. Similar process for creating one tall door that meets an angled ceiling, where they will want to start with a duplicate then add the angle in the 3D editor,

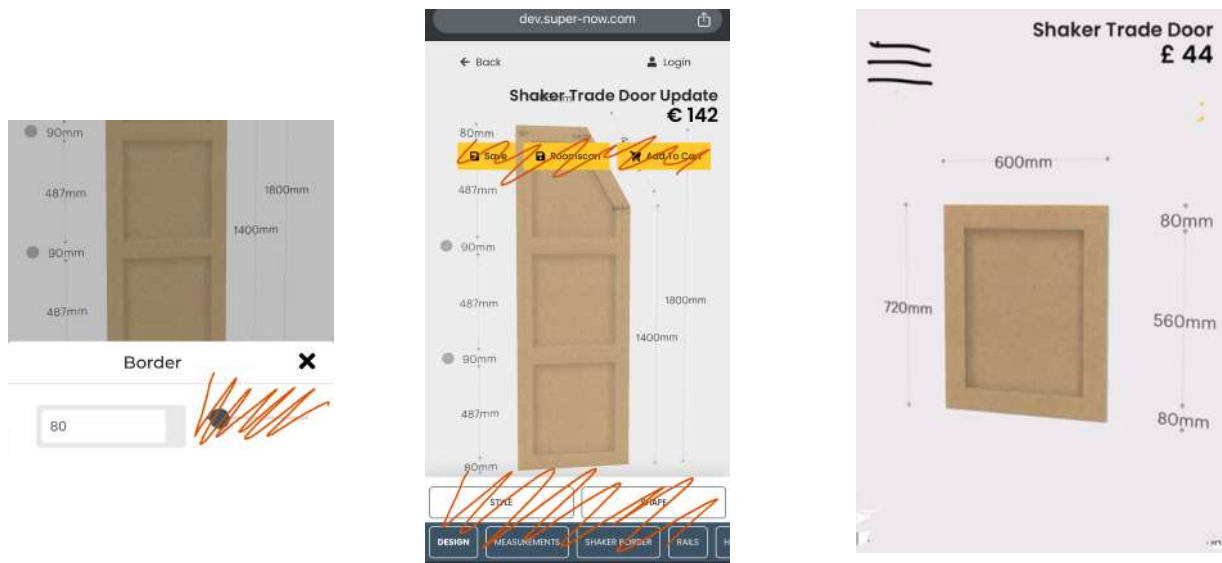


3D Door Interface Improvements

Things to get rid of!

The current mobile menu is just too bitty, too many separate menus with too little information displayed in one place. And the yellow buttons interfere with the 3D model too much on phone and could be tidied up. The room scan feature is not needed at all here. **and please remove all sliders as they are not needed - typing in numbers is better**

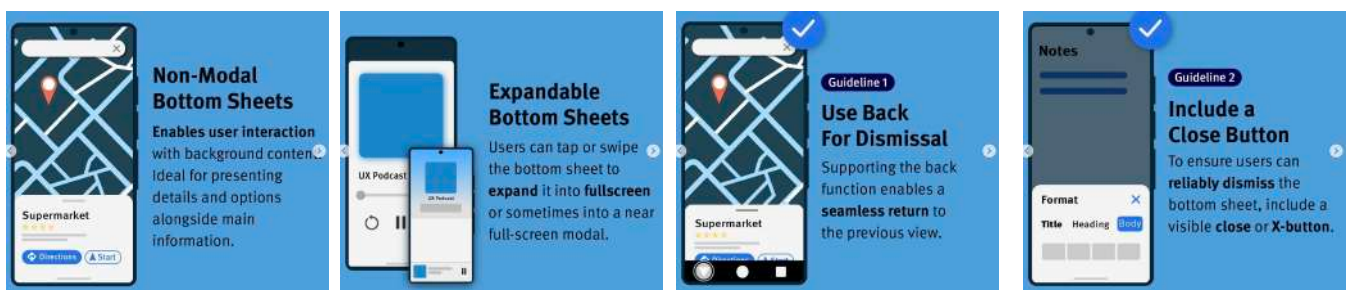
However the title of the door and the price readout should remain at top of the view. Some options could be gathered into a hamburger button menu, perhaps at top left.



Mobile Interface Preferences

The main door editing options should preferably be accessible from bottom of the screen, just in a much better designed menu than before. If possible this menu should be similar in general appearance and usage to what displays on the desktop version as a side bar.

The principles shown below seem appropriate - imagine the map screen is the 3D door, and the bottom sheet is a set of options that could be pulled up to almost cover the 3D door, but you have a strong sense of it still being there, the title and price always remains visible above, and and it is easy to minimise the settings sheet to see the changes on the 3D door. But it is still there peeking up from the bottom of the screen, inviting you to pull it back up again.



Here is a very rough example of how the bottom sheet might look when in general editing mode in 3D door view:

Some settings like the height, width and default border width require a text box to be tapped, then new text to be entered. These settings should be at the top to allow for the keyboard to appear below.

Other settings only require a tap of a box/button to toggle between fixed options, and these can be nearer the bottom as they won't require the keyboard.



The black circles represent 'i' circles - they are inviting the user to tap them to find out more information. **this is an essential*

requirement of the interface - that important information can be communicated at point of need, because we have found that when we try to communicate everything in one place it is too much and users don't read it!!

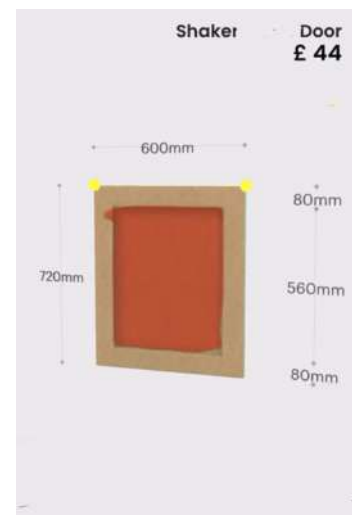
*So for example, at the point when users are toggling between the levels of finish, this is when they will want to know more detail, and so this is where the little information button needs to be, and it will open up another overlay text box to clarify what the options mean.**

More detailed settings should be revealed when specific door parts are selected

The general menu (/bottom sheet) as shown above should be the default one that shows when no specific part of the door has been selected. But if the user taps on the panel of the door, that panel should show up as highlighted, and then the available bottom sheet settings will become panel options. These options will include:

- **Toggle buttons:** 12mm MDF / 19mm Reeded / 18mm Canvas Effect Melamine - and in future, fretwork options and other options
- **A single button marked either 'split' or 'add mid rail'.** This should turn the selected panel into 2 panels, with a rail in between which will be sized according to the 'default border width' already set in the main settings.

Angle settings could perhaps be accessible by tapping on circles at the corners of the door.



Custom border width settings that depart from the default could be accessed by tapping the centre of a specific border, then typing the desired width into a text box. If tapping a mid rail, then there should be an additional field for specifying the height of the rail from the bottom of the door. Similar with spinning the door round then tapping a hinge. Alternatively, these dimensions might be edited by tapping the dimensions that already show on the screen and editing these directly. This might be

more intuitive and could prevent the door being obscured by a bottom sheet. But I am happy to go with whichever option is easier to program and deliver with minimal problems.

My Role

I am working with you to develop this system in return for:

1: Exclusive rights to use the system for manufactured and delivered doors via my business Freebird Interiors LTD within a limited geographical area TBC (provisionally a 50 mile radius from our base in Sheffield, UK - see image - but in reality this may be defined more by county lines as we firm up partner agreements.)

2: A percentage of the sales using this door ordering system from any other providers who adopt it in the UK and worldwide. I am already having conversations with potential UK partners, and can attract more interest when I publicise it via YouTube and Instagram, and at the upcoming shows. The percentage should reflect both the development work and the marketing work that I can bring to the table, while keeping the pricing attractive to users.

As agreed in the video call with Thierry and Paolo on 4th April, we will discuss and then formalise a written agreement on percentages and other details during Week Commencing 13th May 2024, to put a clear agreement in place prior to the weekend of 18/19th when the system will be publicised at the exhibitions. I will want to understand the mechanism by which any agreed percentage of sales is recorded, collected and passed on to me.

I suggest that we agree an initial limited period during which I run the system as sole manufacturer of this style of door, to allow us to fully error check it and for me to establish it's credibility and gather user reviews, and photos of completed projects ordered using the system. Then we will be in a stronger position to roll it out to carefully chosen partners, or however you like.

I will be open to creating video content on how to use it if there is sufficient incentive, and doing this should minimise on-boarding time for us all.

All of the above is of course just my own suggestions, I appreciate you allowing me this input in your own system, and my intention is only to work with you for mutual benefit for us all!

Thanks,

Alastair

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8th April 2024